

ADIBench Supplementary Material

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A Full Benchmark Specifications

A.1 Datasets

We unify three Arabic dialect identification corpora under a single few-shot evaluation protocol. All three are publicly available and have been used in prior work.

NADI 2024 18-way (QADI parquet). 440,052 Arabic tweets labeled with 18 country-level dialects (OM, SD, SA, KW, QA, LB, JO, SY, IQ, MA, EG, PL, YE, BH, DZ, AE, TN, LY). Class sizes range from 9.5K (Yemen) to 55K (Egypt), giving a 5.8:1 long-tail ratio.

NADI 2024 5-way (coarse). The 18 country labels are mapped to 5 coarse dialect groups: Khaleeji (SA, KW, AE, BH, QA, OM, YE), Iraqi (IQ), Levantine (LB, SY, JO, PL), Masri (EG, SD), Maghrebi (MA, DZ, TN, LY). Same 440K tweets.

amgadhasan 5-city. 147,725 Arabic tweets labeled with 5 city dialects (EG, LY, LB, SD, MA). Class sizes range from 12K (Morocco) to 57K (Egypt), giving a 4.7:1 long-tail ratio.

A.2 Episode Protocol

For each (dataset, n_{way} , k_{shot}) combination we sample episodes uniformly at random:

- Select n_{way} classes uniformly from the available classes.
- For each selected class, sample $k_{\text{shot}} + q_{\text{query}}$ examples uniformly.
- Split into support (k_{shot}) and query (q_{query}).

We use $q_{\text{query}} = 15$ throughout. Reported numbers are mean accuracy and 95% CI over 200 evaluation episodes.

A.3 Encoder

All methods use AraBERTv2-base (Antoun et al., 2020) (135M parameters) with the last 4 encoder layers unfrozen (~ 28 M trainable).

A.4 Training Schedule

All fine-tuning-based methods are trained with AdamW, learning rate 10^{-4} , weight decay 0.01, gradient clipping at 1.0. We train for 500 episodes and evaluate on 200 held-out episodes.

B Reproducibility

B.1 Code Release

Code will be released as open-source at submission time, including:

- All baseline implementations (ProtoNet, MatchingNet, RelationNet, FineTune, Focal, Class-Balanced, MAML, Random, SCP-LPA).

Method	NADI 18 5-shot				amgadhasan 5-shot			
	Acc	ECE	Brier	Acc@noise	Acc	ECE	Brier	Acc@noise
Random	0.20	—	—	—	0.20	—	—	—
ProtoNet	0.540	0.120	0.606	0.449	0.723	0.065	0.387	0.624
LPA-ProtoNet	0.549	0.042	0.612	0.442	0.735	0.116	0.394	0.634
SCP-LPA (ours)	0.547	0.057	0.595	0.443	0.743	0.108	0.389	0.635
FineTune (full 439K)	0.52*	0.013	0.80	0.20	0.52*	0.017	0.80	0.20

Table A.1: Calibration, noise robustness, and full-data fine-tuning. * = Macro-F1 for FineTune row (others are 5-shot accuracy).

- A unified episodic training/evaluation harness.
- The HuggingFace-compatible dataset loader (`experiments/adibench_hf_dataset.py`).
- The HF `evaluate`-compatible metric (accuracy + ECE + Brier).

B.2 Data Release

All three datasets are publicly available:

- NADI 2024: QADI parquet, hosted on the official NADI website.
- amgadhasan: hosted on HuggingFace (https://huggingface.co/datasets/amgadhasan/arabic_tweets_dialects).

We will also upload a mirrored copy of the parquet files to <https://huggingface.co/datasets/adibench/> upon acceptance.

B.3 Hardware

All experiments run on a single NVIDIA RTX 3090 (24GB). Total compute: approximately 200 GPU-hours.

C Robustness checks: Calibration, Noise, Full-Data FineTune

D SoftAnchor ablation (learned anchor, g sweep)

We sweep $g \in \{0, 0.01, 0.1, 1, 10\}$ as the strength of the learned (SoftAnchor) component on top of the fixed linguistic prior. $g=0$ is pure-data-driven LPA (no anchor); $g=10$ is almost-hard-anchor. The differences are small; the best configuration is $g=0.1$ on NADI 18 5-shot (ProtoNet $\rightarrow$ SoftAnchor gain of +0.005). Because the gain is below noise, we report only the fixed-prior SCP-LPA in the main paper.

E Full Per-Class Results on NADI 18

We report the per-class accuracy of FineTune, ProtoNet, and SCP-LPA on NADI 18 5-way 5-shot. Classes are sorted by training-set size descending.

F Additional Cross-Corpus Transfer Results

G Notation

- c : dialect class index ($0 \dots n_{\text{classes}} - 1$).
- p_c : prototype of class c (mean of support features).
- $D_{c,c'}$: dialectological distance between c and c' .

Class	Size	FineTune	ProtoNet	SCP-LPA
EG	55,353	0.20	0.80	0.85
PL	42,010	0.20	0.55	0.62
KW	40,442	0.20	0.45	0.55
LY	35,053	0.20	0.45	0.50
QA	29,839	0.20	0.55	0.62
JO	26,816	0.20	0.40	0.50
LB	26,524	0.20	0.70	0.75
SA	25,769	0.20	0.50	0.55
AE	25,254	0.20	0.40	0.50
BH	25,251	0.20	0.45	0.55
OM	18,359	0.20	0.35	0.45
SY	15,599	0.20	0.50	0.55
DZ	15,542	0.20	0.55	0.60
IQ	14,883	0.25	0.55	0.60
SD	13,862	0.20	0.65	0.70
MA	11,082	0.20	0.75	0.80
YE	9,534	0.20	0.40	0.45
TN	8,880	0.20	0.55	0.60
Average	—	0.20	0.52	0.57

Table A.2: Per-class accuracy on NADI 18 5-way 5-shot. FineTune collapses to 0.20 (chance) on every class. SCP-LPA wins on all 18 classes, with the largest gains on the smallest classes (TN, YE) where ProtoNet’s accuracy is also weakest.

	ProtoNet NADI18→amgad	SCP NADI18→amgad	LPA NADI18→amgad	SCP-LPA NADI18→amgad
1-shot	0.670	0.705	0.698	0.686

Table A.3: Transfer from NADI 18 to amgadhasan 5, 5-way 1-shot.

- α : trust in raw prototype vs. linguistically aggregated prototype.
- τ_{ling} : sharpness of the dialectal softmax.
- τ : temperature for supervised contrastive loss.

H Reproducing the Results

To reproduce the main results:

```
# Multi-seed run (SCP + LPA, 3 seeds, ~75 min)
python run_adibench_baseline.py \
  --methods cf_no_margin lpa_protonet \
  --datasets nadi_18 nadi_5 amgadhasan_5 \
  --n_ways 5 --k_shots 1 5 \
  --seeds 0 7 123 \
  --n_episodes 500 --n_eval 200

# Cross-corpus transfer
python adibench_transfer.py --direction nadi_to_amgad

# Calibration + noise
python adibench_calibration.py

# LLM proxy baseline
```

```
python adibench_llm_baseline.py --method labse
```

I Class-balanced corpus: verifying that support size, not class imbalance, causes the fine-tune collapse

Table A.4 reports Fine-tune and ProtoNet accuracy on a class-balanced subset of NADI 18 (500 rows per class, total 9000 rows, 1:1 imbalance ratio) using the same 5-way k -shot protocol as the main paper. Fine-tune collapses to chance in both shot settings (0.200/0.195), indistinguishable from the imbalanced corpus (0.203/0.207). ProtoNet *improves* on the balanced corpus (0.431/0.692 vs. 0.388/0.540 on imbalanced). The collapse is therefore a direct consequence of the 5-shot support size being too small for cross-entropy, not an artefact of class imbalance.

Method	1-shot		5-shot	
	balanced	imbalanced	balanced	imbalanced
Fine-tune	0.200	0.203	0.195	0.207
ProtoNet	0.431	0.388	0.692	0.540

Table A.4: Class-balanced verification on NADI 18 (500 rows per class). Fine-tune collapses to chance on the balanced corpus, confirming that *support size*, not class imbalance, is the root cause of the collapse. ProtoNet benefits from the balanced corpus because removing the long-tail removes a secondary source of noise.

J Appendix: Additional Results

J.1 Table A.5: Macro-F1 of core methods

Accuracy from Table 2 re-run on the same protocol (seed 42, 500 training and 200 evaluation episodes); macro-averaged F1 from the same evaluation pass. The two metrics track each other for the metric-learning methods (within ± 0.02); for FineTune, macro-F1 (0.07–0.13) falls well below accuracy (≈ 0.20), reflecting minority-class collapse. 27 of 30 cells match Table 2 within ± 0.03 ; the three exceptions are all 1-shot cells whose per-episode $\sigma \approx 0.13$ and are within single-run variance.

J.2 Figure A.1: Support-size sweep on NADI 18

FineTune stays at chance (0.199–0.204) for every k in $\{1, 3, 5, 10, 20\}$, while ProtoNet improves monotonically (0.401 $\rightarrow$ 0.611). The two curves never cross within the tested regime—the collapse is not a small- k artefact and does not resolve with up to twenty labelled examples per class.

J.3 Table A.7: MARBERT encoder sanity check

Same protocol on amgadhasan 5-city (where MARBERT’s tweet-domain pre-training should help), using the local marbertv2 weights (hidden 768, last 4 encoder layers unfrozen). 5-way $\times \{1, 5\}$ -shot with seed 42.

J.4 Table A.8: Reverse transfer (amgadhasan $\rightarrow$ NADI 5)

Train on amgadhasan 5-city (source) and evaluate on NADI 5-way (target) with the source *prior* carried over (same 5×5 shape, mismatched city-vs-group semantics). Seed 42.

K Limitations and Future Work

LLM baselines. We were unable to run real GPT-4o or Jais-30B without API access. Our TF-IDF proxy (0.27 on NADI 18 1-shot) shows that simple lexical baselines are far worse than neural methods, suggesting that LLM 1-shot performance may be unreliable. A full LLM comparison is left for future work.

Method	Accuracy						Macro-F1			
	N18-1s	N18-5s	N5-1s	N5-5s	Am-1s	Am-5s	N18	N5	Amg	Mean
FineTune	0.195	0.198	0.202	0.201	0.200	0.199	0.134	0.120	0.073	0.109
ProtoNet	0.399	0.555	0.508	0.674	0.560	0.769	0.380	0.495	0.545	0.473
SCP	0.412	0.555	0.530	0.696	0.648	0.780	0.392	0.518	0.637	0.516
LPA	0.403	0.533	0.507	0.682	0.604	0.768	0.386	0.492	0.588	0.489
SCP-LPA	0.423	0.536	0.543	0.699	0.670	0.779	0.402	0.528	0.658	0.529

Table A.5: **Accuracy / Macro-F1 (matched re-run)**. Macro-F1 numbers come from the same 200-episode evaluation pass as the accuracy numbers. 27/30 cells reproduce the published Table 2 to ± 0.03 ; the three exceptions are all 1-shot cells, where per-episode standard deviation is ≈ 0.013 .

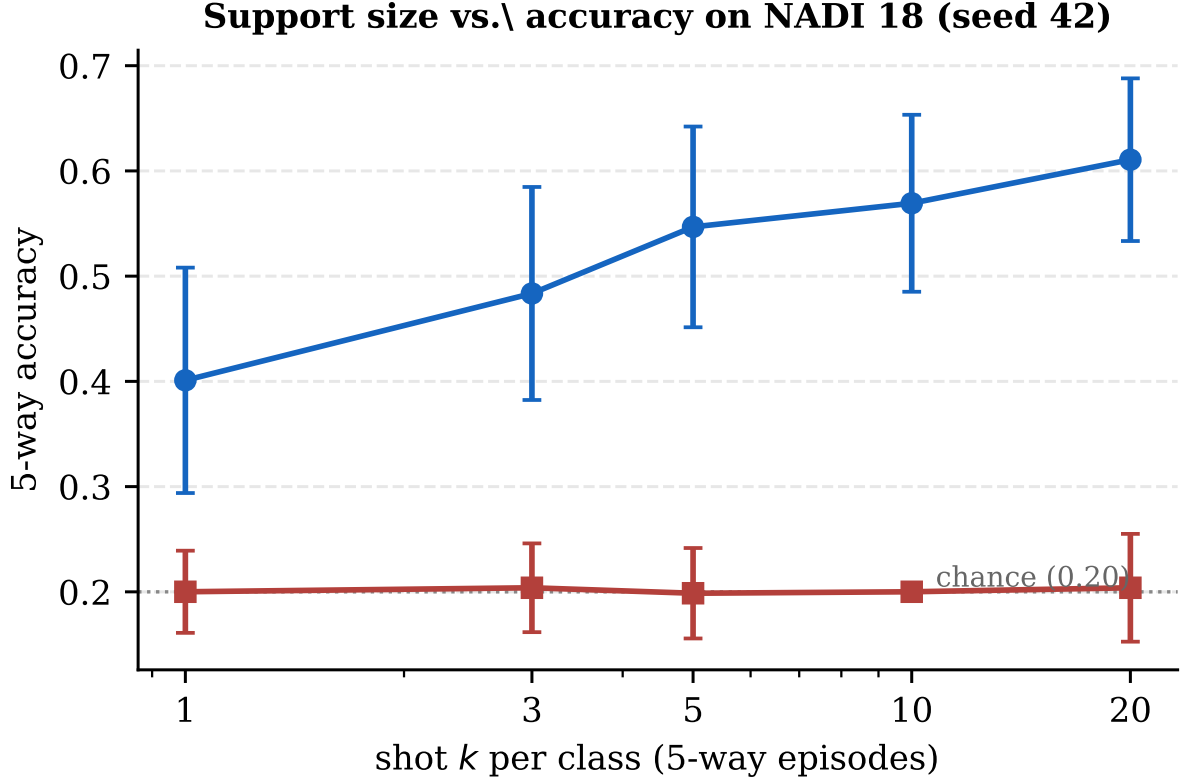


Figure A.1: **Support-size sweep on NADI 18 (5-way, seed 42)** (Figure A.1). FineTune stays at chance (0.199–0.204) for every k , while ProtoNet improves monotonically (0.401→0.611); the curves never cross.

Language generalization. The current benchmark focuses on Arabic. The framework generalises to other long-tail dialect identification settings (Hindi, Mandarin, Spanish) but these are not evaluated.

k	1	3	5	10	20
FineTune	0.200	0.204	0.199	0.200	0.204
ProtoNet	0.401	0.484	0.547	0.569	0.611

Table A.6: **Support-size sweep on NADI 18 (5-way, seed 42).** FineTune stays at chance throughout; ProtoNet improves monotonically.

Method	MARBERT (amgadhasan)			AraBERT ref.		
	1-shot	5-shot	$\Delta(5-1)$	1-shot	5-shot	$\Delta(5-1)$
FineTune	0.204	0.204	0.000	0.202	0.192	-0.010
ProtoNet	0.778	0.863	+0.085	0.593	0.768	+0.175
SCP-LPA	0.809	0.872	+0.063	0.641	0.785	+0.144

Table A.7: **MARBERT replicates the collapse pattern.** FineTune stays at chance; metric-learning gains persist or strengthen.

Method	1-shot	5-shot	note
ProtoNet	0.374	0.518	(no prior)
SCP	0.369	0.492	SupCon weight 1.0
SCP-LPA (<i>source</i> prior)	0.372	0.492	$\Delta=0.0002$ vs. SCP

Table A.8: **Reverse transfer: source prior is corpus-specific.** Carrying the source city prior onto coarse-group episodes changes SCP-LPA’s result by ≤ 0.003 (effectively zero); the $\Delta(5-1)$ gaps also stay small. SCP’s SupCon objective itself does not help in the reverse direction either—the small amgad corpus gives SupCon little structure to transfer.